Paleomagnetism and magnetic mineralogy of the Nahant gabbro and tonalite, eastern Massachusetts

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Paleomagnetic analyses of the Nahant gabbro and an associated tonalite of eastern Massachusetts reveal five separate populations of paleomagnetic directions. The gabbro and tonalite, exposed in the northernmost portion of the Boston Basin, were selected for this study because of the lack of significant macroscopic evidence of penetrative metamorphism or deformation. The tonalite has not been radiometrically dated; however, an Ordovician date has been reported for the gabbro. Detailed petrographic data indicate that the five populations of paleomagnetic directions obtained for the gabbro and tonalite correspond to five styles of magnetic mineral alteration.

A mean alteration index was calculated for each site in the Nahant gabbro and tonalite. There is an apparent relationship between magnetite alteration and normalized intensity at a given demagnetization level, with normalized intensity decreasing as magnetite alteration increases. This relationship illustrates the interdependence between paleomagnetic properties and magnetic minerals. Alteration of these minerals is probably associated with periods of magnetic overprinting.

Autometasomatism is believed to have occurred in the gabbro. Fluid flushing initiated plagioclase decomposition, liberating Ca + Al + Si to form Al-rich (4−10 wt.% Al_2O_3) titanite (sphene, $CaTiSiO_5$). Whether the metasomatism was deuteric or the result of postdeuteric hydrothermal activity has not been determined.

The Nahant gabbro poles are interpreted as representing either a deuteric or a metasomatic chemical remanent magnetization (CRM). The tonalite pole is interpreted as a deuteric CRM. The gabbro and tonalite poles are similar to a number of poles from Ordovician to Devonian intrusives in the Acadia terrane, including the St. Stephen and St. George plutons.

Petrographic examination of polished sections from the Devonian St. Stephen and St. George plutons reveals evidence for alteration of primary magnetic mineral assemblages. Similarities exist between these alterations and alterations observed in the Nahant suite, most notably, the formation of titanite from Fe−Ti oxides. The repeated determination of similar poles on intrusive rocks in the Acadia terrane suggests there may be some tectonic or geomagnetic significance to poles from intrusives; however, without more data pertaining to the timing of metasomatic events observed in the Nahant suite and the St. Stephen and St. George plutons, the age and significance of these poles cannot be interpreted with certainty.

Les analyses paléomagnétiques du gabbro de Nahant et d'une tonalite associée de l'est du Massachusetts révèlent cinq familles distinctes de directions paléomagnétiques. Le gabbro et la tonalite, lesquels affleurent dans la partie la plus au nord du bassin de Boston, ont été choisis pour cette étude parce qu'ils ne présentent aucun signe macroscopique de métamorphisme intense. Il n'existe pas de détermination d'âge radiométrique de la tonalite, cependant on sait que le gabbro est d'âge ordovicien. L'étude pétrographique détaillée indiquent que les cinq familles de directions paléomagnétiques obtenues pour le gabbro et la tonalite correspondent à cinq modes d'altération des minéraux magnétiques.

Un indice moyen d'altération a été calculé pour chaque site analysé dans le gabbro et la tonalite. Une relation apparente existe entre l'altération de la magnétite et l'intensité normalisée à un niveau donné de désaimantation, l'intensité normalisée décroit lorsque l'altération de la magnétite augmente. Cette relation met en lumière l'interdépendance entre les propriétés paléomagnétiques et les minéraux magnétiques. L'altération de ces minéraux est possiblement associée aux périodes de surimpression magnétique.

Il semble qu'il y ait eu autométasomatisme du gabbro. La décomposition des plagioclases a été entamée par l'injection de fluides, libérant Ca + Al + Si pour former une titanite (sphène $CaTiSiO_5$) alumineuse (4−10% par poids d'Al). Il a été impossible de déterminer si le métasomatisme fut d'origine deutéritique ou a résulté d'une activité hydrothermale post deutéritique.

Les pôles du gabbro de Nahant sont considérés comme représentant une aimantation rémanente chimique (ARC), soit de nature deutéritique ou métasomatique. Le pôle de la tonalite est interprétée comme une ARC deutéritique. Les pôles de la tonalite et du gabbro sont analogues à un certain nombre de pôles d'intrusions ordoviciennes et dévoniennes à l'intérieur de la terrane d'Acadia, incluant les plutons de St. Stephen et de St. George.

L'altération des minéraux magnétiques primaires dans les assemblages des plutons dévoniens de St. Stephen et de St. George est révélée par l'étude pétrographique des sections polies. Il existe des similitudes entre ces altérations et celles observées dans la séquence de Nahant, particulièrement en ce qui concerne la formation de la titanite à partir des oxydes de Fe−Ti. La valeur répétée de pôles analogues dans les roches intrusives de la terrane d'Acadia sous-entend que les pôles des intrusifs sont possiblement reliés à quelque événement tectonique ou géomagnétique; cependant, l'âge et la signification de ces pôles ne peut pas être interprété avec certitude sans l'obtention de données supplémentaires sur la chronologie des événements métasomatiques observés dans la séquence de Nahant et dans les plutons de St. Stephen et de St. George.

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Introduction

Acadia is a displaced terrane located on the easternmost margin of the Appalachian orogen in North America and the British Caledonides (Kent and Opdyke 1978). This terrane consists of Upper Precambrian basement overlain by Paleozoic strata containing Early to Middle Cambrian fossils of the Acado-Baltic fauna (Osberg and Skehan 1979). Geological evidence suggests that rocks of Acadia were metamorphosed and deformed several times: during the Late Precambrian, during an episode that occurred after the Middle Cambrian and before the Pennsylvanian, and during a Late Pennsylvanian −

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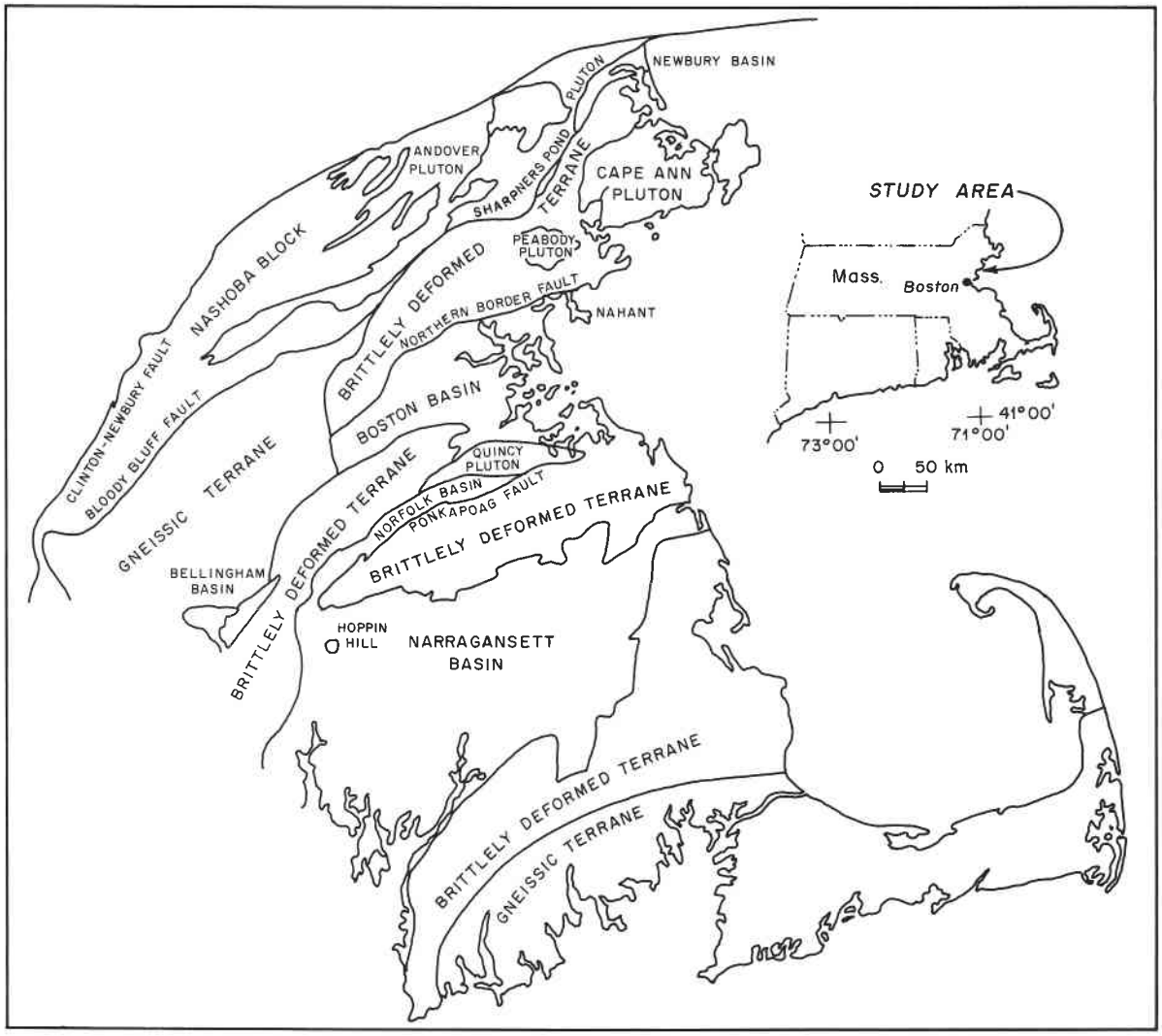


FIG. 1. Tectonic map of eastern Massachusetts east of the Clinton–Newbury fault zone. From Zen (1983).

Permian orogeny. Kent and Opdyke (1978) and Van der Voo *et al.* (1979) suggested that Acadia has undergone approximately 1500 km of post-Devonian left-lateral strike-slip motion, bringing it into its present position with respect to cratonic North America during the Permo-Carboniferous. More recent data (Kent and Opdyke 1984) from the Lower Carboniferous Mauch Chunk Formation, however, suggest that significant latitudinal offset of the Acadia terrane may not have occurred.

This study was designed to obtain a reliable paleomagnetic pole for the Milford–Dedham zone, located in eastern Massachusetts east of the Bloody Bluff fault zone (Fig. 1). The Ordovician Nahant gabbro and a tonalite associated with the gabbro were selected for the study based on their apparent megascopically unmetamorphosed state and the fact that the gabbro has been radiometrically dated by Zartman and Marvin (1971).

Schutts *et al.* (1976) reported a paleomagnetic pole on a Precambrian mafic complex northwest of Boston. This pole was interpreted as Devonian based on biotite–amphibole K–Ar blocking temperatures. Hurley and Shearer (1981) published a paleomagnetic pole on the Proterozoic Z Salem gabbro–diorite, which was also interpreted as Devonian based on the results of Schutts *et al.* (1976).

Geology and age

The Nahant gabbro crops out mainly along the coast of Nahant, approximately 14 km northeast of Boston. The gabbro is intruded into a northeast–southwest-trending syncline in the Lower Cambrian Weymouth Formation. Diorites and tonalites are associated with the gabbro (Fig. 2). Verma (1973) reported veins of the diorite intruding the gabbro on an island off the northeast coast of Nahant. Because no noticeable change in grain size was observed towards the contact, the diorite was not considered a separate intrusion but rather as evidence for multiple intrusive episodes of a single, larger magmatic event. The gabbro is a medium-grained, subhedral to subophitic, greenish gray to dark gray rock containing labradorite and augite, with minor hypersthene, biotite, magnetite, and ilmenite. Epidote has been reported on slickensided fault surfaces (Kaye 1965). Verma (1973) reported evidence for low-grade metamorphism of the gabbro, particularly in the vicinity of small faults. Silicates in the gabbro show evidence of extensive metasomatism and (or) metamorphism. Plagioclase is extensively altered to sericite, and olivine is completely altered to chlorite, talc, sericite, calcite, amphibole, and biotite.

Zartman and Marvin (1971) reported Rb–Sr dates of 493 ± 31 and 461 ± 35 Ma and a K–Ar date of 489 ± 15 Ma, all on magnesian biotite from the gabbro. These data, combined with

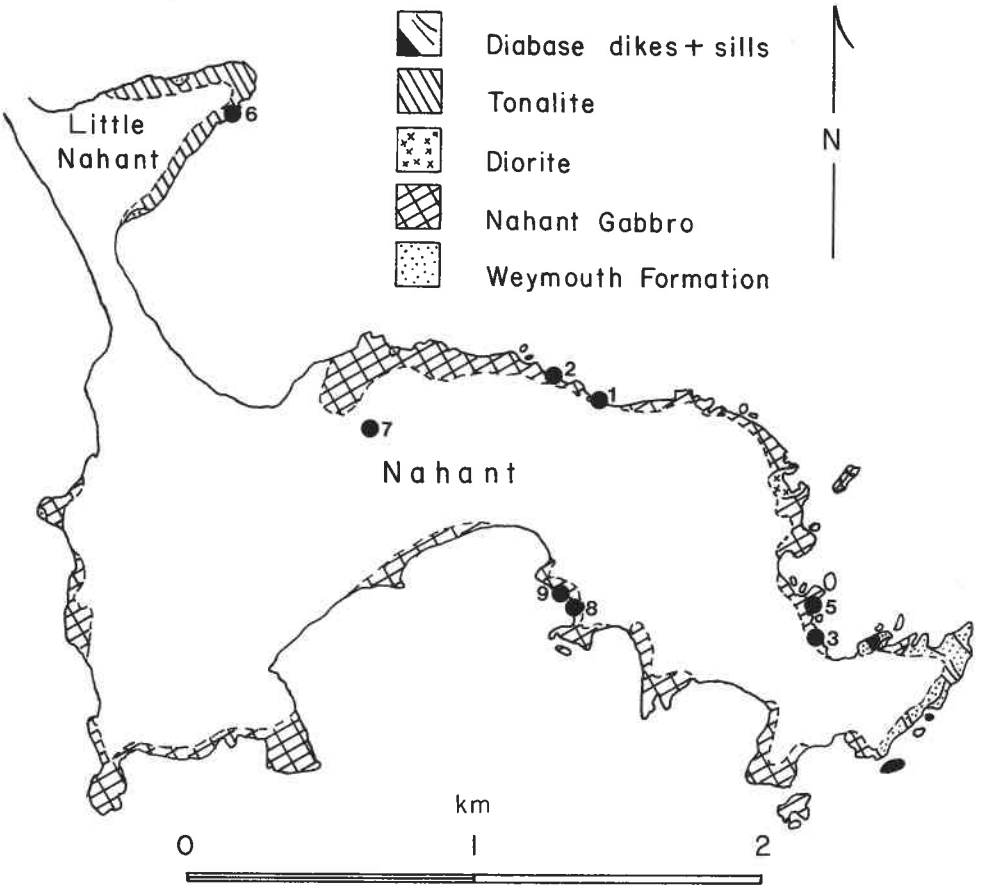


FIG. 2. Generalized geologic map of the exposed geology of Nahant showing the location of paleomagnetic sampling localities. From Verma (1973).

stratigraphic evidence, indicate that emplacement occurred sometime after the Early Cambrian and no later than the Early Ordovician.

The tonalite has a texture similar to that of the main gabbro at Nahant, but the average grain size is smaller (Verma 1973). This rock contains oligoclase, quartz, augite, biotite, magnetite, ilmenite, and chalcopyrite. The tonalite has not been dated. It was considered by Verma (1973) as being associated with the gabbro, although it has not been found to intrude the gabbro, nor is it intruded by the gabbro.

Diabase dikes intrude both the main gabbro at Nahant and the tonalite on Little Nahant. Verma (1973) described the dikes as medium-grained porphyritic rocks containing plagioclase (An_{71}), augite, magnetite, and ilmenite. The Nahant dikes have not been dated, although Ross (1984) reported dikes intruding Precambrian to Devonian age rocks in the Milford–Dedham zone having K–Ar dates ranging from 190 ± 6 to 383 ± 23 Ma; however, some dikes may be as old as Precambrian.

Laboratory procedures

Nine sites were sampled at Nahant, seven in the main gabbro (1–3, 5, 7–9), one in the tonalite (6), and one in a diabase dike (4) that intrudes the gabbro (Fig. 2). A total of 62 samples were obtained. Results from the gabbro and tonalite only are presented in this paper.

Paleomagnetic measurements were performed on a Molspin Minspin magnetometer. Stepwise demagnetization was performed on all but one specimen per site using a Schonstedt alternating field (AF) specimen demagnetizer in fields up to 100 mT. The remaining samples were demagnetized using a Schonstedt thermal demagnetization furnace.

Three 2.5 cm diameter polished sections per site were examined using high power (×400–1000) oil-immersion reflected-light microscopy. Properties such as crystal form, grain size, primary intergrowths, and alteration products were noted for each polished section. Volume percents of the opaque minerals were estimated by point counting an average of 250 grains per section. Thin sections representative of the Nahant gabbro and tonalite were also examined for evidence of silicate metamorphism. Electron microbeam analyses of critical intergrowths were performed on an ETEC Autoprobe using a variety of oxide and silicate standards at 15 kV, 0.03 nA, and a beam diameter of 3 μm.

Paleomagnetic properties

Natural remanent magnetization (NRM) intensities for sites 3, 5, 8, and 9 in the gabbro average about 1 A/m. Sites 1, 2, 6, and 7 have lower intensities that are of the order of 0.1 A/m. NRM directions for the gabbro show considerable scatter (Fig. 3), although there is a general tendency for samples to have steep inclinations. Paleomagnetic results from the gabbro and tonalite, after stepwise AF and thermal demagnetization, proved to be heterogeneous. For the majority of specimens from the gabbro, the demagnetization behavior was extremely inconsistent. Figures 4–8 show five typical types of behavior of samples through progressive AF and thermal demagnetization. Beneath each Zijderveld diagram (Zijderveld 1967) is the normalized intensity decay curve for that sample. In a

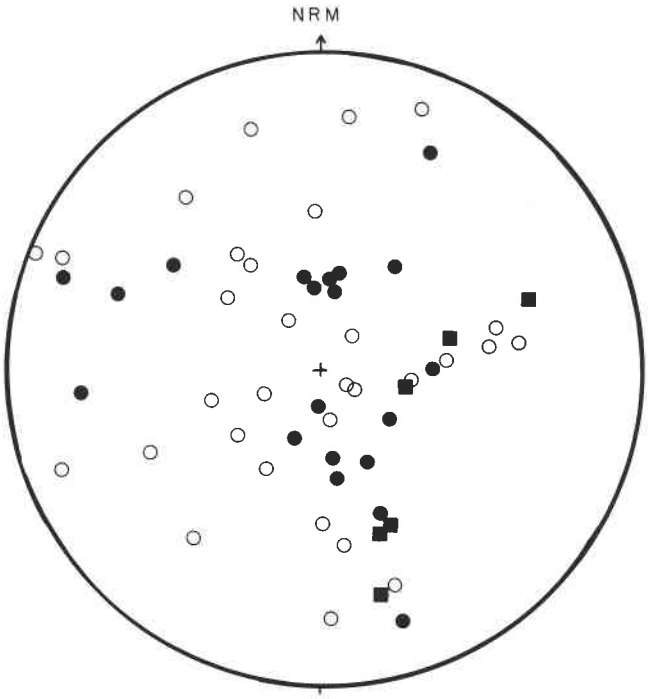


FIG. 3. NRM directions for individual samples of the Nahant gabbro (circles) and tonalite (squares). Closed symbols represent positive inclinations; open symbols represent negative inclinations.

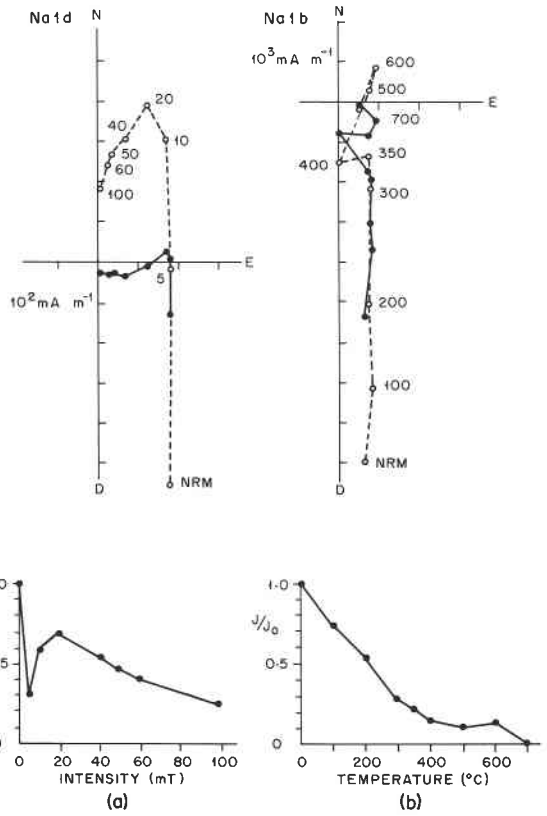


FIG. 4. Zijderveld diagrams from site 1 in the main gabbro for (*a*) AF and (*b*) thermal demagnetization. The magnetization vector in the horizontal plane is shown in solid circles; the vertical plane is shown in open circles. Normalized intensity decay curves are shown below each Zijderveld diagram.

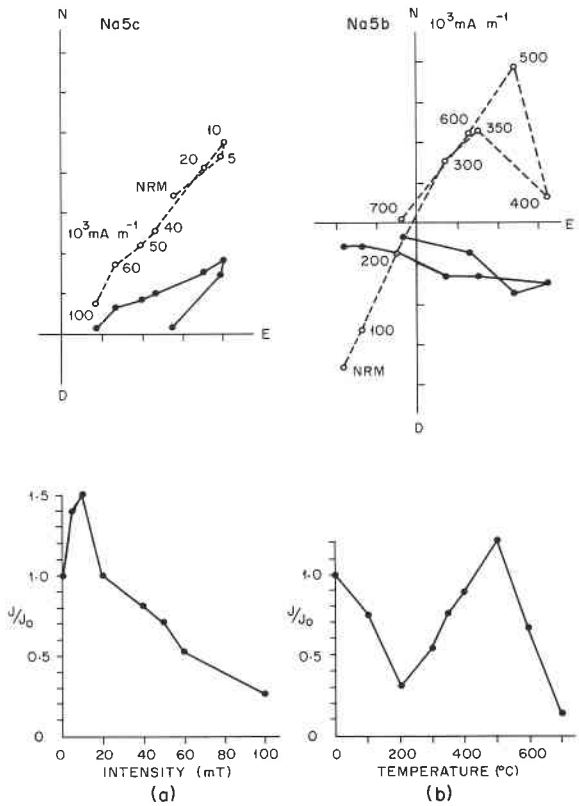


FIG. 5. Zijderveld diagrams from site 5 in the main gabbro for (*a*) AF and (*b*) thermal demagnetization. The magnetization vector in the horizontal plane is shown in solid circles; the vertical plane is shown in open circles. Normalized intensity decay curves are shown below each Zijderveld diagram.

number of samples, the demagnetization behavior is characterized by a final segment that decays in a near linear manner but does not project to the origin (Figs. 4 *a* and *b*, 5*b*, 7*b*, 8*b*). The displacement is probably indicative of higher coercivity or blocking temperature components not completely removed during demagnetization. Another anomalous pattern involves a shift in the direction away from the origin in the final stages of demagnetization, as illustrated in Figs. 7*a* and 8*a*. Figure 5*a* shows a demagnetization behavior with antiparallel directions that appear to decay to the origin. The behavior of sample 5*a* is atypical of the remaining samples from this, or any other site in the gabbro. In the case of the tonalite, site 6, demagnetization was successful in isolating a single component using both AF and thermal demagnetization (Fig. 6 *a* and *b*).

A plot of the cleaned paleomagnetic directions from the Nahant gabbro and tonalite is shown in Fig. 9. For most samples a least-squares fit could not be applied to the demagnetization plots because of erratic behavior. To be consistent, all directions chosen for Fig. 9 were based on demagnetization at 60 mT, 100 mT, or 600°C. Sites 6 and 7 show the best within-site consistency (α_{95} values of 9.7 and 4.5°, respectively). The scatter in the data from the remaining sites reflects the difficulty in isolating a discrete direction in these rocks, particularly in sites 8 and 9, which show at least three components of magnetization during AF treatment.

Magnetic mineralogy

The estimated modal proportions of opaque minerals in the Nahant gabbro are relatively constant, averaging 2–5% by

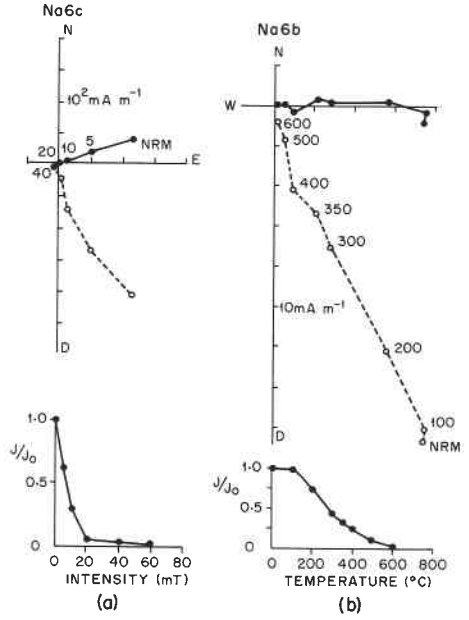


FIG. 6. Zijderveld diagrams from site 6 in the main gabbro for (*a*) AF and (*b*) thermal demagnetization. The magnetization vector in the horizontal plane is shown in solid circles; the vertical plane is shown in open circles. Normalized intensity decay curves are shown below each Zijderveld diagram.

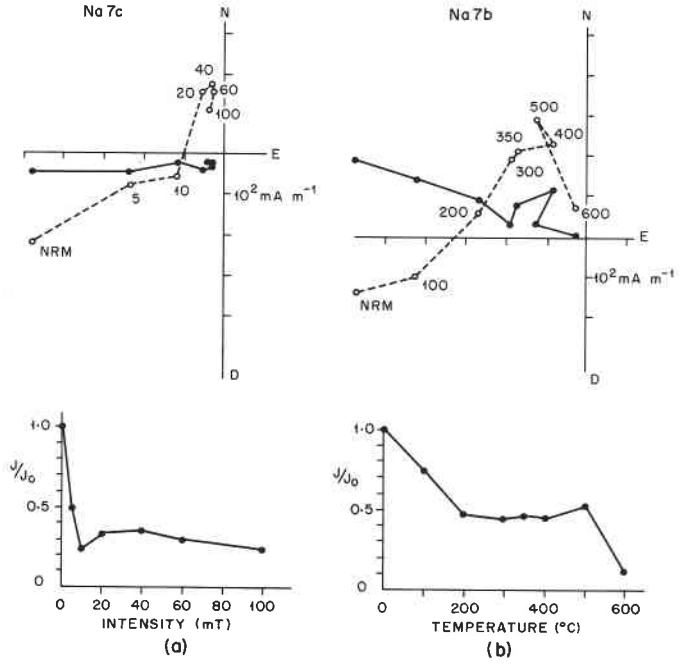


FIG. 7. Zijderveld diagrams from site 7 in the main gabbro for (*a*) AF and (*b*) thermal demagnetization. The magnetization vector in the horizontal plane is shown in solid circles; the vertical plane is shown in open circles. Normalized intensity decay curves are shown below each Zijderveld diagram.

volume. Titanomagnetite and ilmenite are the dominant opaque minerals in all samples, except for site 7 from the main gabbro, which contains large amounts of chalcopyrite. Less abundant oxide phases are hematite, ulvöspinel, pleonaste, rutile, and maghemite. Pyrite is present in all samples from the gabbro but occurs partially altered to goethite in the tonalite (site 6) and the main gabbro at site 8. Chalcopyrite and pyrrhotite with exsolved pentlandite are common sulfides observed in nearly all samples. Covellite (CuS) is only observed in site 7 from the main gabbro. In addition to the primary interstitial oxides (0.1−0.3 mm in diameter) there is also rare magnetite in coarse (0.1 mm) symplectic intergrowths with highly altered silicates at sites 3, 5, 8, and 9, all from the main gabbro.

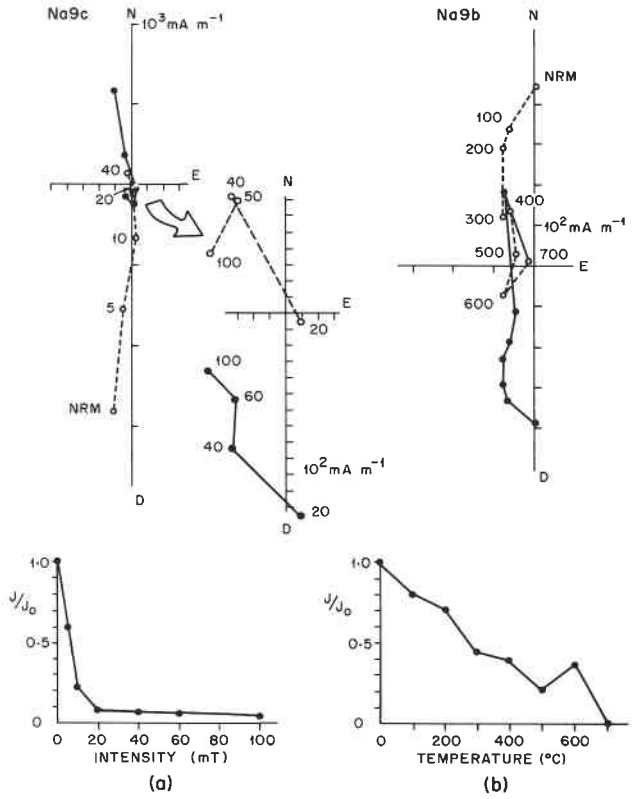


FIG. 8. Zijderveld diagrams from site 9 in the main gabbro for (*a*) AF and (*b*) thermal demagnetization. The magnetization vector in the horizontal plane is shown in solid circles; the vertical plane is shown in open circles. Normalized intensity decay curves are shown below each Zijderveld diagram.

Five distinct styles of alteration are observed in the samples from Nahant (Fig. 10). Sites 1 and 2 contain ilmenite that is unaltered or partially altered to titanite (i.e., sphene, $CaTiSiO_5$), hematite, and rutile. Magnetite is unaltered or slightly altered to titanite. Ilmenite lamellae exsolved from magnetite are commonly altered to titanite, hematite, and rutile. This is designated as the first style of alteration, as illustrated in Fig. 11*a*. The second alteration style is similar to the first, except that the ilmenite is almost completely altered to titanite, hematite, and rutile. Figure 11*b* shows a discrete ilmenite grain that has been extensively altered to titanite, hematite, and rutile; adjacent to this ilmenite is a magnetite grain with {111} lamellae of ilmenite. The magnetite is unaltered, but the ilmenite lamellae have been extensively altered. In both alteration styles 1 and 2 there is a tendency for preferential ilmenite alteration relative to magnetite. Alteration style 2 is observed in sites 3 and 5 from the main gabbro. Alteration style 3 is observed at site 6 in the tonalite on Little Nahant. This site contains exsolved ulvöspinel partially oxidized to ilmenite, intergrown with magnetite altered to maghemite, and disseminated iron staining (Fig. 11*c*). Biotites, which commonly contain titanite lamellae, are occasionally altered to hematite (Fig. 11*d*). The fourth alteration style is observed in samples from site 7 from the main gabbro, which

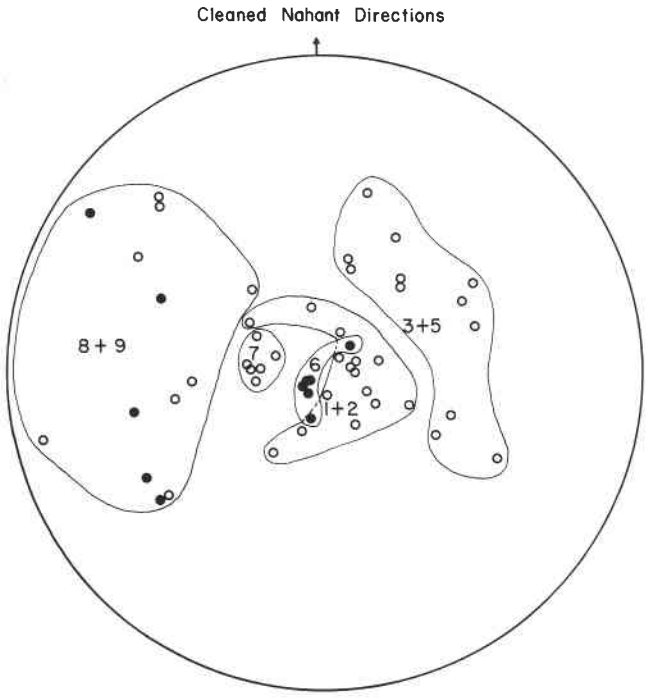


FIG. 9. Paleomagnetic directions for the Nahant gabbro (sites 1–3, 5, 7–9) and tonalite (site 6) after AF and thermal demagnetization. Areas outlined enclose directions of samples from the sites (indicated by site numbers), grouped according to the degree and style of magnetic mineral alterations.

contains large amounts of disseminated chalcopyrite. Magnetite at this site is altered to titanite and (or) "amorphous" iron–titanium oxide; ilmenite is unaltered or slightly altered to titanite, hematite, and rutile. The fifth alteration style is similar to the fourth except that there is only 1 or 2% chalcopyrite in the main gabbro at sites 8 and 9. Figure 11*e* illustrates the tendency for magnetite to be altered instead of ilmenite in sites 7–9 (alteration styles 4 and 5). Magnetite is extensively altered to titanite, with greater amounts of alteration along the edges of the magnetite. An ilmenite lamella, which runs discontinuously along a diagonal of the magnetite grain, is unaltered and has an area of unaltered magnetite immediately adjacent to it. These areas, where magnetite remains unaltered adjacent to ilmenite lamellae, are common in sites 7–9. The phenomenon may be caused by small concentrations of Cr + Al that diffused during ilmenite "exsolution." These elements would act as stabilizers, protecting the magnetite from alteration (Haggerty 1976).

Fine-grained rods of opaque-mineral phases are oriented within silicate minerals, and these may carry a significant, stable remanence because their small sizes approach single-domain or pseudo-single-domain behavior. These opaque rods are too small to be identified microscopically with any certainty even at ×1000 magnification. Their reflectivities suggest an oxide, and they are tentatively identified as either titanomagnetite, or ilmenite, or possibly rutile (Fig. 11*f*).

If the rods are magnetite and are not altered in the same manner or to the same extent as the large, multidomain grains that can be identified, they may account for the steeply upward directions of high coercivities (Figs. 4*a*, 5*a*, 7*a*, 9*a*). However, at least three facts argue against the preservation of these rods and maintenance in a pristine state. Firstly, the very process of oxide exsolution from a silicate is one that effects lattice discontinuities and provides ready access to the encapsulated silicate interior by external fluids or gas components. Secondly, plagioclase is not an effective shield to oxide alteration in the Nahant intrusives because plagioclase decomposition is intimately involved in the "sphenetization" process. Thirdly, the smaller grains have a higher surface/volume ratio than the large, multidomain grains, and kinetically this favors the alteration of finer grained phases over their coarse grained counterparts.

Several processes have led to the development of the assemblages present in the Nahant intrusives. Ilmenite lamellae within titanomagnetite are produced by high-temperature (greater than 600°C) subsolidus oxidation (Buddington and Lindsley 1964). This is observed in samples from the main gabbro. The rutile + hematite assemblage is also indicative of oxidation but coupled with titanite ($CaTiSiO_5$) necessitates that Ca^{2+} and Si^{4+} be added to the dominantly Fe plus Ti mineral oxides. Both discrete primary ilmenites and ilmenites formed by oxidation in host magnetite are replaced by titanite, hematite, and rutile. Compositions determined by electron microbeam analysis show that titanite replacing magnetite contains about 4 wt.% Al_2O_3, in contrast to titanites replacing discrete ilmenites, which contain about 10 wt.% Al_2O_3. Small Al_2O_3 contents may have been present in the original oxides, but the concentrations observed in these rocks are far in excess of values determined for mineral oxides in either mafic or felsic suites (e.g., Higgins and Ribbe 1976). The titanites are low in iron (1–2 wt.%), and removal of FeO must be invoked to satisfy the reaction and the observed assemblages. There is only indirect evidence for secondary FeO precipitation: chlorite is one possible source, disseminated iron staining another, and a third source is the abundant sulfides. This would, therefore, also require the subsequent addition of sulfur to the system. Iron remobilization has been proposed as a mechanism for the formation of Cornwall-type ore deposits from source regions having undergone extensive "sphenitization" (Davidson and Wyllie 1968). Iron remobilization is considered unlikely in the case of the Nahant intrusives, where alteration is sporadic in intensity and localized in distribution. The Ca + Si + Al is most likely derived from plagioclase, which has been extensively altered to sericite in both the gabbro and tonalite.

Relations between alteration styles and magnetic properties

The five distinct alteration styles correlate directly with the five paleomagnetic directions obtained for the gabbro and the tonalite (Table 1). Because both alteration styles 1 and 2 involve ilmenite alteration to titanite, hematite, and rutile, it is possible that these alteration styles are the result of the same metamorphic or metasomatic episode or episodes. Ilmenite is more extensively altered to titanite, hematite, and rutile in sites altered in style 2 than in sites altered in style 1. Overprinting of paleomagnetic directions would probably be more extensive in sites altered in style 2 than in sites altered in style 1.

Sites altered in style 5 have a greater scattering of paleomagnetic directions than sites altered in other styles. One possible explanation for this scatter might involve tectonic rotation. The scatter could also result from the replacement of the magnetite by titanite, leaving behind only small amounts of the strongest remanence carrier, magnetite.

The formation of ulvöspinel in titanomagnetite (coupled with the alterations observed in style 3) results from slow cooling and exsolution (sensu stricto) and is only observed in the tonalite in association with maghemite, a low-temperature (less than 500°C) oxidation product (e.g., Özdemir and Banerjee

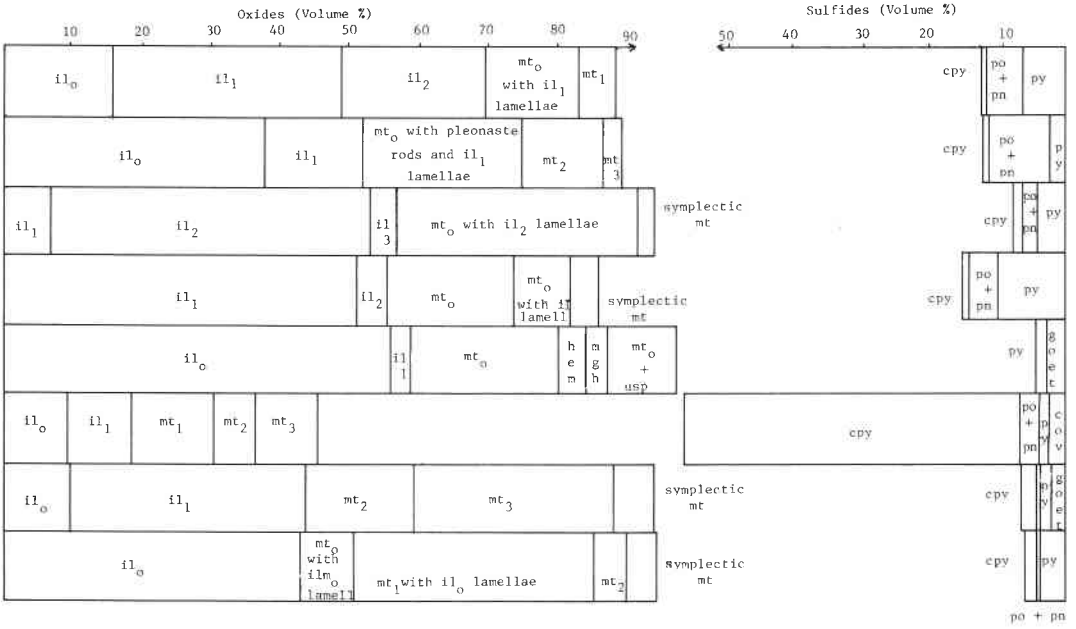


FIG. 10. Volume percent opaque minerals for the eight paleomagnetic sites in the Nahant gabbro (1−3, 5, 7−9) and tonalite (6). Total volume percent of oxides plus sulfides should add to 100% for each row. Oxides and sulfides are shown separately, with oxides on the left and sulfides on the right. Values were obtained by point counting three polished sections per site, with an average of 250 grains counted per section. Abbreviations are: il_0 = unaltered ilmenite; il_1 = ilmenite, <50% altered to titanite, hematite, and rutile; il_2 = ilmenite, >50% altered to titanite, hematite, and rutile; il_3 = ilmenite, 100% altered to titanite, hematite, and rutile; mt_0 = magnetite; mt_1 = magnetite, <50% altered to titanite; mt_2 = magnetite, >50% altered to titanite; mt_3 = magnetite, 100% altered to titanite; hem = hematite; mgh = meghemite; cpy = chalcopyrite; po + pn = pyrrhotite with exsolved pentlandite; py = pyrite; goet = geothite; cov = covellite.

1984). This particular style of alteration could be either deuteric or metasomatic. Maghemite is not thought to be a weathering product, because the tonalite is the only body exhibiting this form of oxidation (Fig. 10). If the alteration is in fact deuteric, then the paleomagnetic direction for this rock may be primary (i.e., deuteric chemical remanent magnetization (CRM) following continuously after thermoremanent magnetization (TRM).

The other four alteration styles (1, 2, 4, 5) are more likely the result of metasomatism. Whether metasomatism was deuteric or the result of later hydrothermal activity is not known and cannot be determined without further isotopic studies of the gabbro.

Watkins and Haggerty (1967) demonstrated that variations in oxidation state exist within single lava flow units. They found that the intensity and stability of magnetization correlate with high Fe−Ti oxide oxidation, with the highest oxidation state occurring at a distance of approximately two thirds from the tops of flows. A modified form of their classification scheme is used here, with a mean alteration index for magnetite and ilmenite calculated for each site in the Nahant gabbro and tonalite using the relationship

$$\text{Alteration index}_M = \frac{(mt_0 * n_0) + (mt_1 * n_1) + (mt_2 * n_2) + (mt_3 * n_3)}{n_0 + n_1 + n_2 + n_3}$$

$$\text{Alteration index}_I = \frac{(il_0 * n_0) + (il_1 * n_1) + (il_2 * n_2) + (il_3 + n_3)}{n_0 + n_1 + n_2 + n_3}$$

where mt_1 represents unaltered magnetite; mt_2, less than 50% altered magnetite; mt_3, greater than 50% altered magnetite; mt_4, completely altered magnetite; ilm_1, unaltered ilmenite; ilm_2, less than 50% altered ilmenite; ilm_3, greater than 50% altered ilmenite; ilm_4, completely altered ilmenite; and n is percentage of grains in each class.

The higher the mean alteration index, the more extensive is the degree of alteration. Figure 12 shows the variations in magnetic properties with the mean alteration numbers at a demagnetization level of 10 mT. There is a relationship for both the mean magnetite and ilmenite indices. Samples with low magnetite alteration indices have high normalized intensities, and those with the highest magnetite alteration indices have the lowest normalized intensities. Because magnetite is strongly magnetic, the intensity of magnetization for a sample should *decrease* if magnetite is destroyed. Hence, extensive alteration lowers the intensity of magnetization of a sample. In contrast, ilmenite is paramagnetic. When altered to titanite, hematite, and rutile, there is a net gain of remanence because secondary hematite is formed, which is weakly magnetic. The more extensive the ilmenite alteration, the greater is the proportion of hematite produced. Larger concentrations of hematite and the consequent relation to magnetic intensity also result because the ilmenite is initially more reduced than coexisting titanomagnetite, making it more susceptible to alteration. Greater proportions of TiO_2 in the ilmenite allow for greater extraction of TiO_2 to form titanite. Once all of the calcium is satisfied by forming titanite, the remaining titanium forms rutile, and large residual concentrations of iron are produced, which form hematite. Sites with extensive ilmenite alteration have experienced little or no magnetite alteration; hence, the high normalized intensities are more likely due to the contribution from the magnetite, which is much more strongly magnetic than hematite. The fact that the mean ilmenite index for site 2 is anomalously low whereas the normalized intensity for this site is high indicates the predominance of the magnetite contribution. Also anomalous is the tonalite: in every case it is very different from the gabbro, and

TABLE 1. Nahant gabbro

Alteration style	D (°)	I (°)	k	α_{95}	n	Pole position Lat. (°)	Pole position Long. (°)
(1) Magnetite unaltered; ilmenite unaltered or partially altered to titanite, hematite, and rutile	143.2	−82.1	17.4	9.1	16	−53.5	93.4
(2) Magnetite unaltered; ilmenite almost completely altered to titanite, hematite, and rutile	64.5	−65.9	6.6	17.5	13	−16.6	70.1
(3) Magnetite and ulvöspinel, with magnetite altered to maghemite and ulvöspinel oxidized to ilmenite; biotite altered to hematite	224.4	82.2	48.6	9.7	6	31.8	277.4
(4) Ilmenite unaltered or slightly altered to titanite, hematite, and rutile; magnetite altered to titanite; large percentages of fine-grained, disseminated chalcopyrite	−86.9	−74.8	217.9	4.5	6	−34.7	144.4
(5) Ilmenite unaltered or slightly altered to titanite, hematite, and rutile; magnetite altered to titanite	−82.8	−16.4	3.1	28.6	13	−0.1	182.4

NOTES: The five alteration styles are listed with the associated paleomagnetic directions; statistical data for each of the average directions are given as well as the corresponding paleomagnetic pole; D and I are the declination and inclination, respectively; k is Fisher's precision parameter; α_{95} is the radius of the cone of 95% confidence about the mean direction; n is the number of samples averaged.

its temporal relationship to the gabbro may need to be reconsidered.

Discussion of paleopole positions

The complex and variable alteration styles observed in the Nahant gabbro can be correlated with changes in magnetic intensity. Because magnetic intensity variations were also observed as a result of *deuteric* oxidation in Icelandic lava flows, it cannot be determined whether the Nahant gabbro poles represent a deuteric or metasomatic CRM. The paleomagnetic poles for the gabbro, therefore, cannot be interpreted with any certainty as primary or secondary based on the presently available petrologic and radiometric data. In the case of the tonalite, however, the style of alteration is most likely deuteric; hence, the tonalite pole could be interpreted as a meaningful direction representative of the Earth's field at the time the tonalite cooled.

Poles for the five populations from the Nahant gabbro and the tonalite are shown in a paleopole plot (Fig. 13), along with poles derived from Ordovician to Devonian rock units for North America and Avalon. The Nahant gabbro and tonalite poles are highly dispersed and plot in intermediate to high northern latitudes. Pole positions similar to the Nahant poles were obtained from the St. Stephen and St. George plutons (Roy *et al.* 1979), the Mount Peyton batholith (Lapointe 1979), the Bonavista and Deadman's Bay dikes (Murthy 1983*a*,*b*), and the Cape Breton granites (Rao *et al.* 1981), all of Devonian age. The Arden pluton (metamorphosed in the Ordovician) pole also resembles the Nahant poles (Rao and Van der Voo 1980).

Poles that plot in this position have led to controversy regarding the reliability of intrusive rocks for paleomagnetic studies. Kent and Opdyke (1980) suggested that results from intrusive rocks may not be reliable records of the paleomagnetic field in all cases. They cited the lack of correction for tectonic tilting in intrusive bodies as well as the presence of multiple remanences as possible sources of divergence. The possibility that the Nahant gabbro has experienced postemplacement tectonic rotation cannot be ruled out as a factor contributing to the divergence of the Nahant poles. The petrological evidence outlined in this paper, however, indicates that the magnetic minerals have been affected during metasomatism and that this is primarily the cause of divergency.

If the tonalite is Ordovician, then it is difficult to understand how it escaped the metasomatism so evident in the gabbro. The tonalite has not been dated, however; evidence indicating that it is associated with the Nahant gabbro is inconclusive, and hence it is possible that the tonalite is younger than the Nahant gabbro. This conclusion is appealing, as the intrusion of the tonalite could have triggered the metasomatism that affected the gabbro.

The repeated determination of similar poles from plutonic rocks of the Acadia terrane suggests there may be some tectonic or geomagnetic significance to poles obtained on intrusives. Without an age determination for the tonalite and an estimate on the timing of the gabbro metasomatism, it is necessary to suggest a number of interpretations for the gabbro and tonalite poles.

If the tonalite is Ordovician in age, then the tonalite pole could be representative of the Ordovician geomagnetic field; in this event, gabbro poles 1, 2, 4, and possibly 5, because of their proximity to the tonalite pole, could be the result of either an Ordovician deuteric CRM carried by the secondary oxide assemblages described in Table 2 or an Ordovician TRM carried by the fine-grained opaque phases oriented within silicate minerals. If the tonalite is not Ordovician in age, it is possible that it represents an intrusive episode later in the Paleozoic. The tonalite pole could be representative of the post-Ordovician to Triassic geomagnetic field. Gabbro poles 1, 2, 4, and possibly 5 might then be the result of a metasomatic CRM that may have been triggered by the intrusion of the tonalite. Gabbro pole 5 is only tentatively included in this

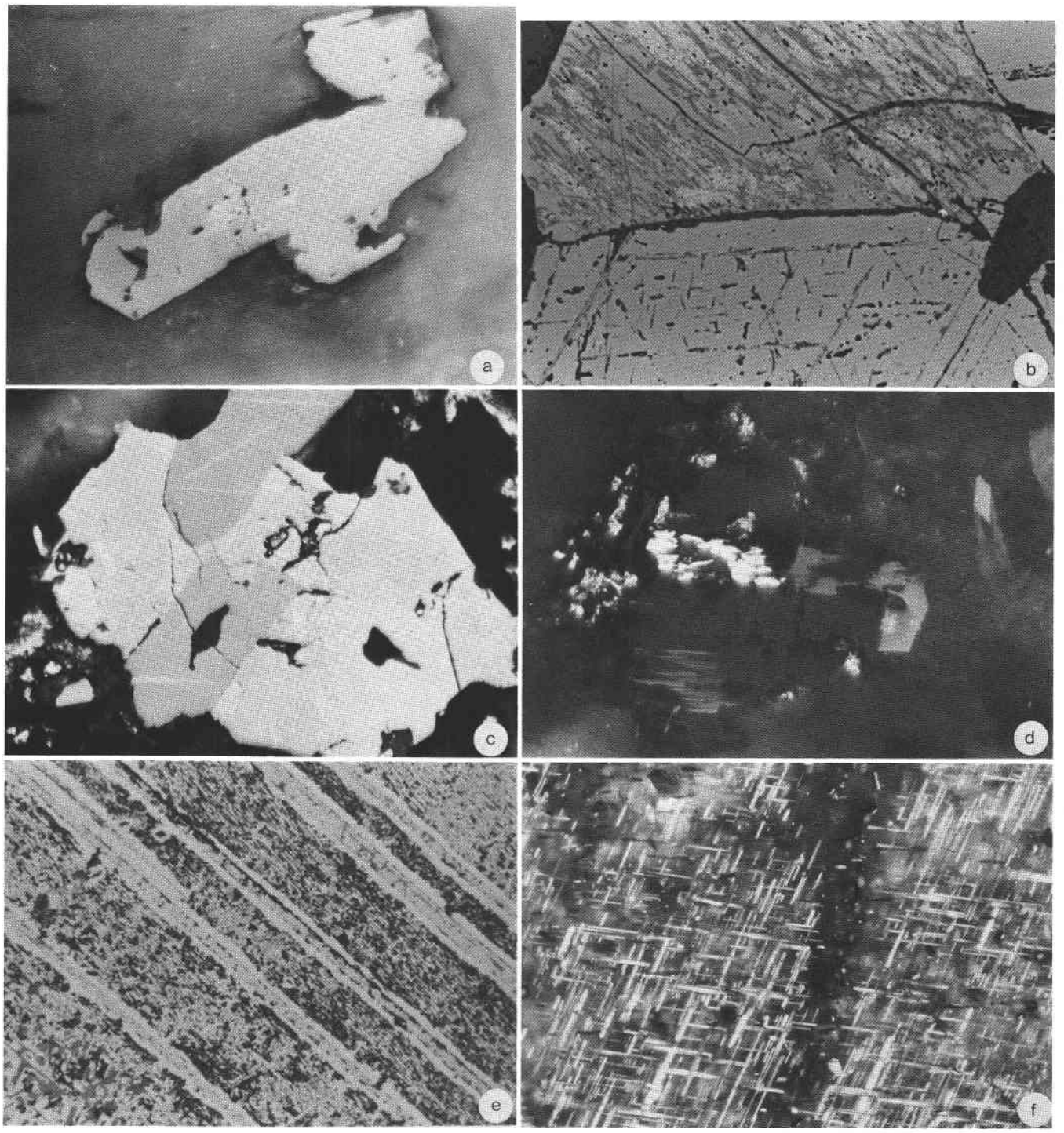


FIG. 11. Reflected-light photomicrographs of magnetic mineral assemblages. (*a*) Photomicrograph of alteration style 1. Width of field = 0.31 mm (oil). Ilmenite is slightly altered to titanite, hematite, and rutile. (*b*) Alteration style 2. Width of field = 0.775 mm (air). Ilmenite has been extensively altered to titanite, hematite, and rutile on the top. Below the ilmenite is an unaltered magnetite grain with pleonaste rods and (111) lamellae of ilmenite, which have also been altered. (*c*) Alteration style 3. Width of field = 0.31 mm (oil). In the center is a large intergrowth of magnetite and ulvöspinel (patchy areas). The magnetite has been altered to maghemite in the cracks. Three large ilmenite grains are in external composite relationship to the magnetite–ulvöspinel intergrowth. (*d*) Biotite altered to hematite. Width of field = 0.31 mm (oil). This is characteristic of alteration style 3. Lamellae of titanite are also present in the biotite. Euhedral zircons can be seen to the right of the biotite grain. These zircons are commonly surrounded by fine-grained disseminated iron staining. (*e*) Alteration styles 4 and 5. Magnetite is extensively altered to titanite. Width of field = 0.19 mm (glycerine). Note the lack of alteration of the magnetite immediately adjacent to the ilmenite lamellae that run discontinuously across a diagonal from the upper left to the lower right of the photograph. (*f*) Rods of an opaque phase. Width of field = 0.31 mm (oil). These rods cannot be positively identified microscopically but are important if they are magnetic because they may carry a very stable remanence as a result of their small size, which approaches single-domain or pseudo-single-domain dimensions.

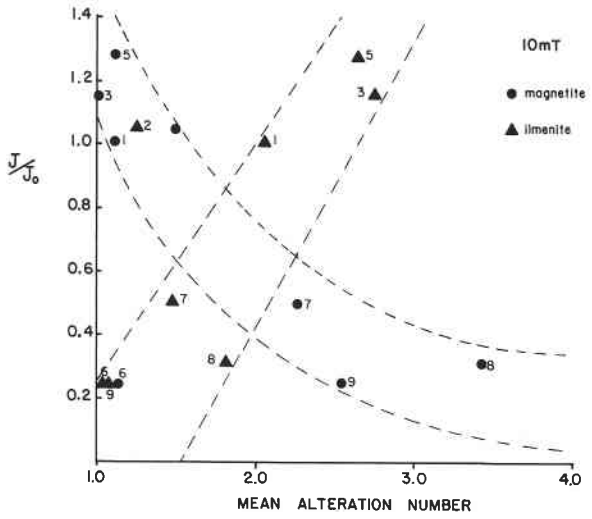


FIG. 12. Mean alteration index for each site in the Nahant gabbro and tonalite (site 6) versus the average normalized intensity for each site at 10 mT. Dashed lines enclose regions containing a majority of ilmenite or magnetite values.

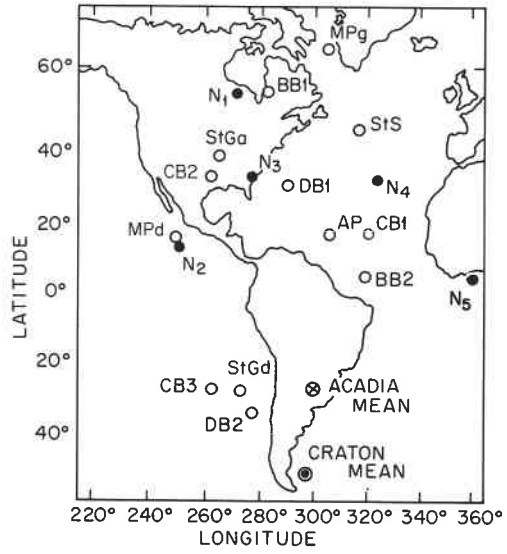


FIG. 13. Paleopole (south) plot for the Nahant gabbro (N_1, N_2, N_4, N_5) and tonalite (N_3). Other poles are: Mount Peyton granite (MPg); Mount Peyton diorite (MPd) (Lapointe 1979); Bonavista Bay dikes (BB1, BB2) (Murthy 1983*b*); Deadman's Bay dikes (DB1, DB2) (Murthy 1983*a*); St. George pluton (StGa, StGd), and St. Stephen pluton (StS) (Roy *et al.* 1979); Cape Breton granites (CB1, CB2, CB3) (Rao *et al.* 1981); Arden pluton (AP) (Rao and Van der Voo 1980); Acadia and craton means (Van der Voo and Scotese 1981).

interpretation because the data for sites having this pole were highly dispersed (see Table 1; Fig 8 *a* and *b*). In addition, these sites experienced the most extensive magnetite alteration (Fig. 10).

Preliminary results from the examination of polished sections from the St. Stephen and St. George pluton (on loan from J. L. Roy) indicate a degree and style of alteration that differ to some extent from those observed in the Nahant suite. Ultramafic samples from the St. Stephen pluton contain primary chrome-spinel and magnetite, as well as secondary magnetite blebs set in a predominantly sericite matrix. Hematite occurs as lamellae within biotite. Samples from the St. George pluton contain both magnetite–ilmenite composite grains and magnetite pseudographically intergrown with biotite. Minor "sphenitization" of these magnetites has occurred. Pyrite is either unaltered or completely altered to a FeO · OH polymorph (goethite, lepidocrocite, or akaganeite). The controlling factor for this alteration appears to be grain size or surface/volume ratio. The more fine-grained pyrites experience a greater degree of alteration because of their higher surface/volume ratio. Hematite needles concentrated along cracks and grain boundaries in magnetite were only observed in sections from the St. George pluton subjected to thermal demagnetization to 640°C, and these may be an artifact of the heating experiments.

More data are clearly necessary to determine whether a relationship exists between the assemblages observed in the St. Stephen and St. George plutons and the paleomagnetic poles for these intrusives. A thorough petrological study may provide valuable evidence concerning the nature and timing of the metasomatism observed.

Conclusions

This study illustrates the ease with which a delicate high-temperature magnetic system may be modified by superimposed chemical magnetization processes. Postconsolidation fluid flushing or autometasomatism has taken place; titanite has formed from Fe–Ti oxide minerals through reaction with Ca + Si + Al liberated from the breakdown of plagioclase. Thermal metamorphic conditions may not be required to affect the paleomagnetic signature of a rock. Therefore, the absence of a metamorphic fabric or the mineral assemblages in a rock are not sufficient criteria to determine the degree to which that unit has retained its primary remanence.

Correlation of paleomagnetic data with Fe–Ti oxide alteration indicates that a relationship exists between the degree of alteration and the magnetic intensity of a particular sample. These results appear promising, but additional data are required to better define the relationship between complex and variable alterations and the magnetic properties of these rocks.

The Nahant paleomagnetic poles, when compared with Ordovician and Devonian poles from Acadia and North America, are most similar to poles obtained on Devonian intrusive rocks, including the St. Stephen and St. George plutons. Alterations observed in the Nahant gabbro could be either deuteric or metasomatic. The gabbro poles, therefore, represent either a deuteric or metasomatic CRM. Alterations observed in the tonalite are deuteric; hence, the tonalite pole most likely represents a deuteric CRM. Preliminary reflected-light data from the St. George and St. Stephen plutons indicates that these rocks have experienced Fe–Ti oxide and Fe sulfide alterations. The timing of these alterations is not certain, nor is it understood to what nature and extent these metasomatic assemblages relate to the multiple remanences observed in these units.

Reevaluation of all the Devonian intrusive poles after the examination of the opaque-mineral assemblages would provide valuable information that could lead to a better understanding of intrusive rocks and their reliability as recorders of the Earth's magnetic field.

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